Lab Report 10 – TORQUE

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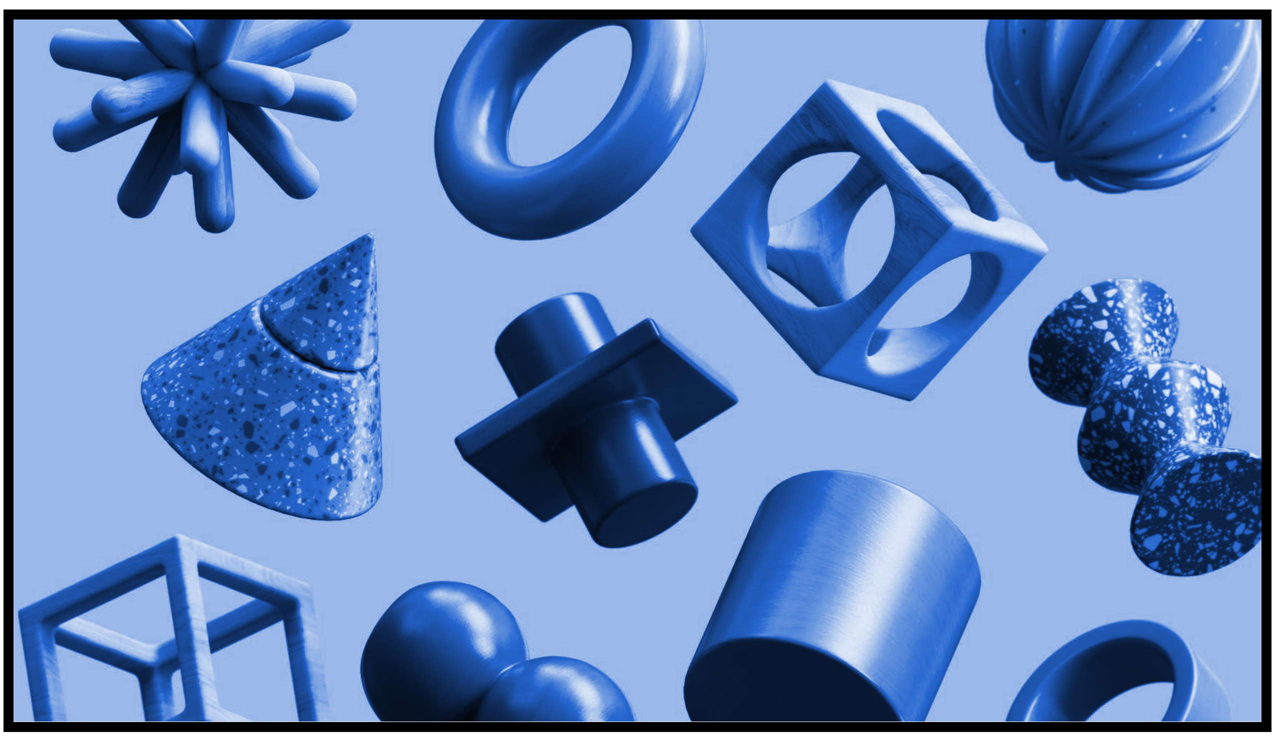


Objectives

To test predictions of the laws of angular motion.

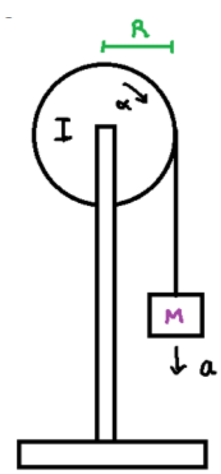
Equipments

The list of equipments used in this experiment is:

- Hanging mass
- Torque apparatus
- Rotary motion sensor
- Computer interface
- Beam balance
- Meter stick

Theory and Equations

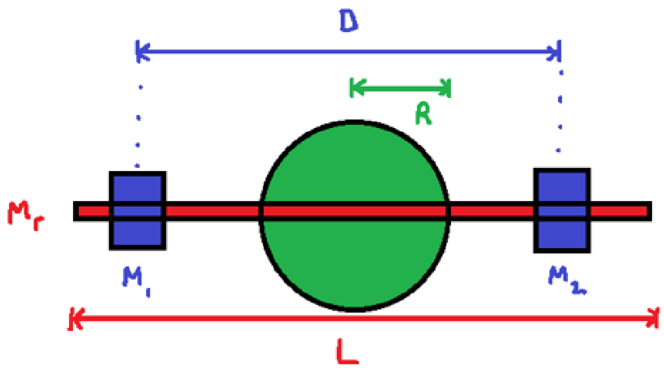
1. Torque Theory and Formula



Consider a pulley with a non-zero moment of inertia I and radius R. If we attach a hanging mass M to a string wound around the pulley, then we can predict the angular acceleration when the mass is released:

$$\alpha = \frac{(g/R)}{1+\frac{I}{mR^2}} \qquad \text{(Eq. 1)}$$

Consider the case where the pulley has negligible mass, but the moment of inertia is non-negligible due to an attached rod and 2 small masses:



Radius of pulley = R

Distance between small masses = D

Small masses have values M_1 and M_2

Length of Rod = L

Mass of Rod = M_r

The moment of inertia for the pulley system:

$$I = \frac{1}{12} M_r L^2 + (M_1 + M_2) \frac{D^2}{4} \qquad \text{(Eq. 2)}$$

2. **Percentage Error**

The percentage error is calculated based on the experimentally determined measure (exp) and the accepted value (acc). And there are 2 types of accepted values which are theoretical value which is predicted from theory and empirical value which is collected from a reliable source.

$$\text{Percentage Error} = \left| \frac{(exp) - (acc)}{(acc)} \right| * 100\% \qquad \text{units: \%}$$

Summary of Procedures

To begin, we establish a stable surface using the specified apparatus. Then, the mass values of the rod, mass 1, mass 2, and the length of the rod (L) are measured. Next, the apparatus is assembled as illustrated in the images provided, incorporating a 50g hanging mass. Initially, the two masses are placed near the ends of the rod in equilibrium, with the maximum D value recorded. The pulley is then rotated to shorten the length of the string. When the pulley is released, the masses are allowed to fall, causing the pulley to rotate as the computer interface records the angular acceleration (α). This process is repeated, adjusting the position of the two masses approximately 1 cm closer to the center. After the masses are re-equilibrated, the D value is measured and the computer interface is used to determine the new angular acceleration (α). This process is continued until the masses cannot be repositioned any closer to the center.

Data and Observations

Mass 1 and Mass 2 ($m_1 + m_2$) (units: kg)	0.151
Mass of Rod (m_r) (units: kg)	0.0277
Length of rod (L) (units: m)	0.381
Total hanging mass (m) (units: kg)	0.0550
Acc Gravity Acceleration (g) (units: m/s^2)	9.800
Pulley Radius (R) (units: m)	0.0300

ID	D (units: cm)	D (units: m)	exp α (units: rad/s^2)
1	36.0	0.360	2.55
2	34.0	0.340	2.77
3	32.0	0.320	3.20
4	30.0	0.300	3.38
5	28.0	0.280	4.03
6	26.0	0.260	4.62
7	24.0	0.240	5.27
8	22.0	0.220	6.30
9	20.0	0.200	7.17
10	18.0	0.180	8.33
11	16.0	0.160	10.2
12	14.0	0.140	12.3
13	12.0	0.120	14.9
14	10.0	0.100	17.9
15	8.00	0.0800	22.5

Data Analysis

Sample Calculation:

In case 1, given D = 0.360m and measured exp α = 2.55 rad/s^2, we can calculate:

Moment of Inertia = I = $\frac{1}{12} m_r L^2 + (m_1 + m_2)\frac{D^2}{4} =$

$$= \frac{1}{12}(0.0277kg)(0.381m)^2 + (0.151kg)\frac{(0.360m)^2}{4}$$

$$= 0.00523 \text{ kg}.m^2$$

$$\text{acc } \alpha = \frac{(g/R)}{1 + \frac{I}{mR^2}} = \frac{(9.800m/s^2)/(0.0300m)}{1 + \frac{0.0052\ kg.m^2}{(0.0550kg)(0.0300m)^2}} = 3.06\ rad/s^2$$

ID	D (m)	exp α (rad/s^2)	Moment of Inertia (kg·m²)	acc α (rad/s^2)
1	0.360	2.55	0.00523	3.06
2	0.340	2.77	0.00470	3.41
3	0.320	3.20	0.00420	3.80
4	0.300	3.38	0.00373	4.28
5	0.280	4.03	0.00329	4.84
6	0.260	4.62	0.00289	5.51
7	0.240	5.27	0.00251	6.32
8	0.220	6.30	0.00216	7.31
9	0.200	7.17	0.00185	8.53
10	0.180	8.33	0.00156	10.1
11	0.160	10.2	0.00130	12.0
12	0.140	12.3	0.00107	14.4
13	0.120	14.9	0.000879	17.4
14	0.100	17.9	0.000713	21.2
15	0.0800	22.5	0.000577	25.8

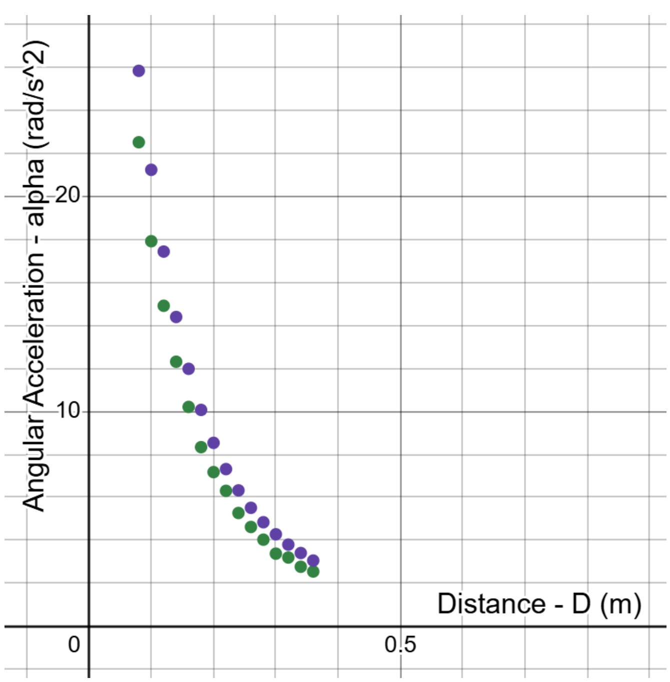
Results

Sample Calculation:

For case 1 of id = 1, we have:

$$\text{Percentage Error} = \left|\frac{(exp)-(acc)}{(acc)}\right| * 100\% = \left|\frac{2.55\,rad/s^2 - 3.06\,rad/s^2}{3.06\,rad/s^2}\right| * 100\% = 16.8\ \%$$

ID	D (m)	exp α (rad/s^2)	acc α (rad/s^2)	Percentage error (%)
1	0.360	2.55	3.06	16.8
2	0.340	2.77	3.41	18.7
3	0.320	3.20	3.80	15.9
4	0.300	3.38	4.28	20.9
5	0.280	4.03	4.84	16.7
6	0.260	4.62	5.51	16.1
7	0.240	5.27	6.32	16.6
8	0.220	6.30	7.31	13.8
9	0.200	7.17	8.53	16.0
10	0.180	8.33	10.1	17.2
11	0.160	10.2	12.0	14.8
12	0.140	12.3	14.4	14.5
13	0.120	14.9	17.4	14.5
14	0.100	17.9	21.2	15.6
15	0.0800	22.5	25.8	12.9



Notation:

- y_1: $exp\ \alpha\ (rad/s^2)$
- y_2: $acc\ \alpha\ (rad/s2)$

Discussion / Conclusion

From the graph of D versus exp α and acc α above, we can notice firstly that for a given value of D, we always seem to get a higher value of acc α in comparison to the exp α. It might come from human error in measuring the masses and length, or from the computer interface.

Interestingly, also from the plot of D versus exp α and acc α above, we can see that both have the same shape of the rational function. This confirms that the theoretical formula is correct. That also means the value of α corresponds to the ratio of $\frac{1}{D^n}$.

In the future, to increase the accuracy of experiments, we may need to use a more detailed scaler.

Post-Lab Question

From Eq.1, we have: $$\alpha = \frac{(g/R)}{1+\frac{I}{mR^2}}$$

$$\alpha = \frac{(g/R)}{1+\frac{I}{mR^2}} = \frac{(g/R)}{\frac{mR^2+I}{mR^2}} = \frac{\frac{g}{R}*(mR^2)}{mR^2+I} = \frac{gmR}{mR^2+I}$$

From Eq.2, we get: $$I = \frac{1}{12}m_rL^2 + (m_1+m_2)\frac{D^2}{4}$$

Replace the formula of I into the modified formula of α above, then:

$$\alpha = \frac{gmR}{mR^2+I}$$

$$\alpha = \frac{gmR}{mR^2+(\frac{1}{12}m_rL^2+(m_1+m_2)\frac{D^2}{4})}$$

$$\alpha = \frac{1}{\frac{mR^2+(\frac{1}{12}m_rL^2+(m_1+m_2)\frac{D^2}{4})}{gmR}}$$

$$\alpha = \frac{1}{\frac{mR^2}{gmR}+\frac{\frac{1}{12}m_rL^2+(m_1+m_2)\frac{D^2}{4}}{gmR}}$$

$$\alpha = \frac{1}{\left(\frac{R}{g} + \frac{\frac{1}{12} m_r L^2}{gmR}\right) + \frac{(m_1 + m_2)}{4gmR} D^2}$$

With b = $\frac{R}{g} + \frac{\frac{1}{12} m_r L^2}{gmR}$; a = $\frac{m_1 + m_2}{4gmR}$

So, now the form of α looks like:

$$y = \frac{1}{b + ax^2} \qquad \text{(Eq.3)}$$

Now, we calculate the value of b and a.

$$b = \frac{R}{g} + \frac{\frac{1}{12} m_r L^2}{gmR} = \frac{(0.0300\ m)}{(9.800\ m/s^2)} + \frac{\frac{1}{12}(0.0277 kg)(0.381 m)^2}{(9.800\ m/s^2)\ (0.0550 kg)(0.0300\ m)} \approx 0.0238$$

$$a = \frac{m_1 + m_2}{4gmR} = \frac{(0.151\ kg)}{(4)(9.800\ m/s^2)\ (0.0550 kg)(0.0300\ m)} \approx 2.34$$

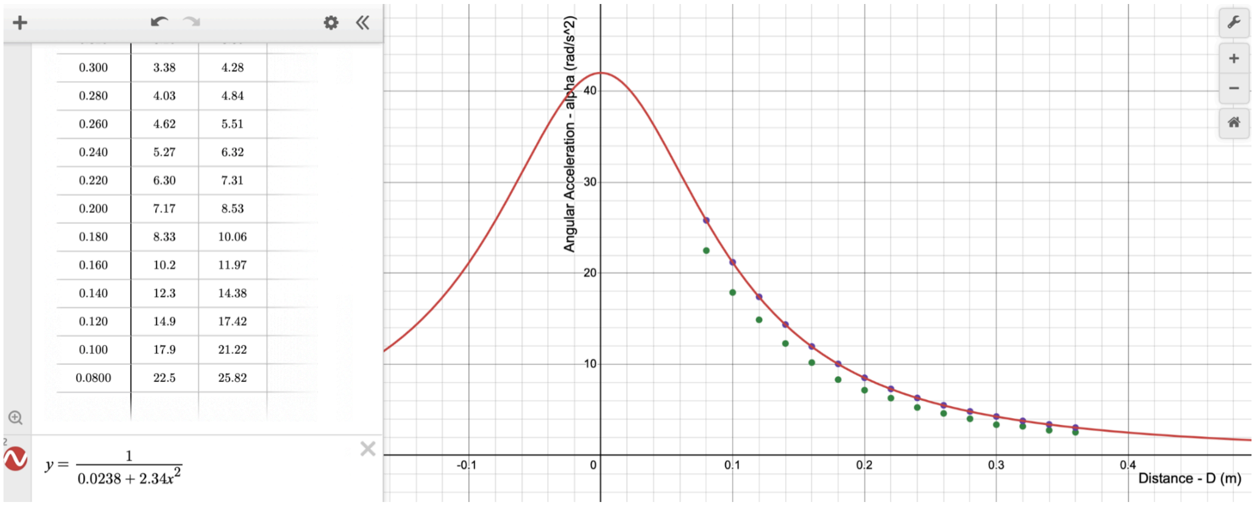
Replace the value of a and b into the Eq. 3 above, we have:

$$y = \frac{1}{0.0238 + 2.34x^2}$$

Thus, we have the function of α as a function of D:

$$\boldsymbol{\alpha} = \frac{1}{0.0238 + 2.34D^2}$$

We can check the function α by plotting the graph below. We can see that it is in perfect alignment with the accepted values of α that we are looking for.



0.300	3.38	4.28
0.280	4.03	4.84
0.260	4.62	5.51
0.240	5.27	6.32
0.220	6.30	7.31
0.200	7.17	8.53
0.180	8.33	10.06
0.160	10.2	11.97
0.140	12.3	14.38
0.120	14.9	17.42
0.100	17.9	21.22
0.0800	22.5	25.82

Notation:

- y_1: $exp\ \alpha\ (rad/s^2)$
- y_2: $acc\ \alpha\ (rad/s2)$